Forward Deployed Engineer (FDE)

Where Business Strategy Meets AI Engineering in 2026

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AGENDA

What We'll Cover Today

6 Short Chapters on Career, AI & Forward Deployed Engineering

01

Self-Introduction

Study Background → SWE

02

What is an FDE?

PLTR to 2026 Boom

03

FDE in the AI Era

10x Force Multiplier

04

How I Study

Gemini + Learning by Doing

05

Why I Chose FDE

Not EM or Core SWE

06

Your Path & Q&A

Fit Check & Open Floor

From Classical Humanities to AI Engineering

Bridging ancient text analysis with modern software development

ACADEMIC BACKGROUND

This is what I studied

Chinese Language & Classical Philology



- **Chinese Major**

Deep focus on classical texts and linguistic structure.

- **Textual Analysis**

Deciphering ancient bamboo manuscripts.

- **Language = Logic**

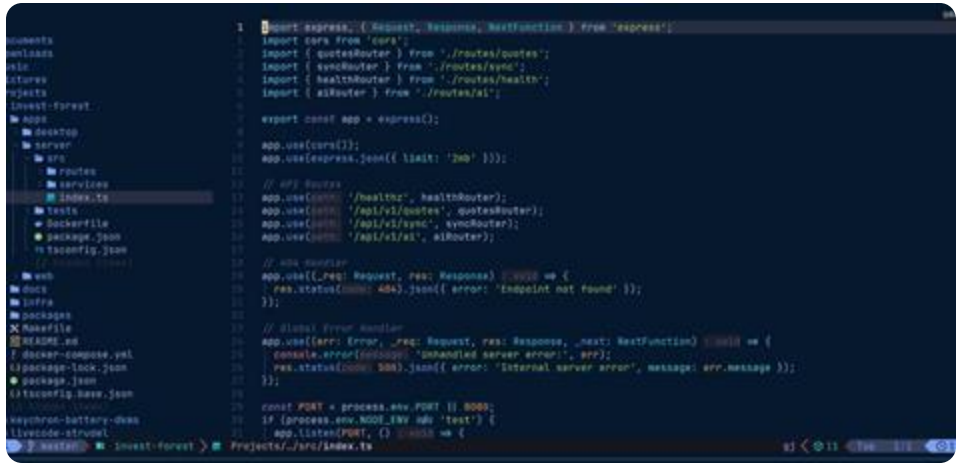
Pattern extraction from unstructured data.

CURRENT PRACTICE

This is what I do

Full-Stack & Forward Deployed Engineering

- **Software Engineering:** TypeScript, API Gateways & Cloud Infra
- **AI Integration:** Building AI routers, microservices & production systems



About Me — Eddie Chan (Part 2)

Combining Engineering Rigor with Business Strategy

03 | MY EDUCATION

Keep Studying

Engineering + Business

- **School of Engineering (eCommerce)**
- **School of Business(Business Administration / Consulting)**
- **Bilingual:** Code + Strategy

04 | MY OBSESSION

Solving Real Problems

Business Pain & Complex Puzzles

★ **Real Business Problems**

★ **High-Stakes Puzzles**

★ **Code is the Tool,** Not the Goal

01 | THE TURNING POINT

The AlphaGo Story (2016)

Why a Chinese Language Student Decided to Learn AI & Code

MARCH 2016 • ALPHAGO VS. LEE SEDOL

Move 37

1-in-10,000 Creative Move

AI Solves Deep Puzzles

01 | THE SPARK

Watched AlphaGo beat Lee Sedol

02 | THE PIVOT

Self-learned Python, ML & Data Science

03 | THE LESSON

Starting major ≠ Your career ceiling

What is a Forward Deployed Engineer (FDE)?

The Most Hands-On Role in Enterprise Tech

CORE DEFINITION

Embed With Clients • Solve Real Pain • Ship Production AI in Days

ORIGIN

Born at Palantir (PLTR)

ENVIRONMENT

Embedded On-Site with Clients

DELIVERABLE

Working Production Code

02 | 2026 MARKET TREND

Why is FDE the Hottest Job in 2026?

From Palantir's Secret Weapon to the #1 Growth Role in AI

TREND 01

AI Labs Scaling FDE

Frontier AI Expansion

- OpenAI, Anthropic & Google
- Hiring FDEs to win enterprise

TREND 02

80% of PoCs Stall

Models Fast; Clients Lag

- Demos are easy to build
- Production deployment is hard

TREND 03

Outcomes > Syntax

AI Writes Boilerplate Code

- Raw coding syntax is cheap
- Business ROI is the scarce moat

02 | THE PROBLEM

What Problem Are FDEs Actually Solving?

Why Traditional Enterprise Software & AI Projects Fail

FAILURE 01

Wrong Problem

Chatbots ≠ Business ROI

- **Clients misdiagnose pain**
- **Build demos without P&L value**

FAILURE 02

Messy Reality

Siloed Data & Security

- **Fragmented legacy ERPs**
- **Strict IAM & compliance walls**

FAILURE 03

Broken Handoffs

Sales → Consultant → SWE

- **4-layer telephone game**
- **Zero real-world user context**

02 | DAY-TO-DAY

What Do FDEs Actually Do?

Owning the Full Loop From Executive Whiteboard to Production

01 | DIAGNOSE

Sit With Users

- Shadow real workflows
- Find true bottleneck

02 | PROTOTYPE

Build in Days

- Code on real data
- Skip 6-month specs

03 | DEPLOY

Ship to Prod

- Pass security & RBAC
- Add evals & guardrails

04 | PRODUCTIZE

Scale Pattern

- Extract core modules
- Feed back to platform

02 | THE FORMULA

FDE = Business × Technology

Why My Two Master's Degrees (Engineering + Business) Compound Here

SCHOOL OF BUSINESS

Business Fluency

- P&L & ROI Framing
- C-Suite Alignment
- Workflow Adoption

×

SCHOOL OF ENGINEERING

Deep Technology

- Full-Stack & Cloud
- AI Agents & RAG
- Data Science Rigor

=

THE FDE SWEET SPOT

10x Business Impact

- Boardroom at 10 AM
- Production Code at 4 PM
- Zero Translation Loss

In the AI Era, FDE Is a 10x Force Multiplier

1 FDE + AI Agents = An Entire Engineering & Consulting Squad

BEFORE AI AGENTS

The Old 6-Person Squad

Slow, Heavy & Expensive

- ✕ **1 PM + 1 BA** writing specs
- ✕ **3 Backend/Frontend SWEs**
- ✕ **3–6 Months** to ship v1

2026 AI ERA

1 FDE + AI Coding Agents

10x Faster End-to-End Delivery

- ★ **AI Agents** handle boilerplate
- ★ **1 Human FDE** owns architecture
- ★ **3–6 Days** to production ROI

The Last-Mile Bridge & AI-Native Builder

Turning Raw Frontier Models Into Live Business Impact

ROLE 02 | THE LAST-MILE BRIDGE

Frontier LLMs → Real ROI

Closing the Enterprise Adoption Gap

- **Raw Models** are a commodity API
- **FDEs Build** data, auth & evals
- **Result:** Measurable P&L impact

ROLE 03 | AI-NATIVE BUILDER

Live Co-Building

SWE + Product Manager + Founder

- **Skip 100-Page** waterfall specs
- **10 AM:** Hear executive pain point
- **4 PM:** Demo live working system

04 | HOW I STUDY

How to Become an FDE? How Do I Study?

3 Core Habits That Took Me From Chinese Language to FDE

HABIT 01

Learning by Doing

Build First, Theory Second

- **Escape 'tutorial hell'**
- **Ship real end-to-end apps**

HABIT 02

Leverage AI

Gemini & NotebookLM

- **24/7 Socratic AI tutor**
- **Master new domains in 48h**

HABIT 03

Startup Mindset

Act Like a Founder

- **Never say 'not my job'**
- **Ship 80% v1 today**

I Am Not Smart, I Am Well Trained

Consistency & High-Volume Deliberate Practice Over Innate Ability

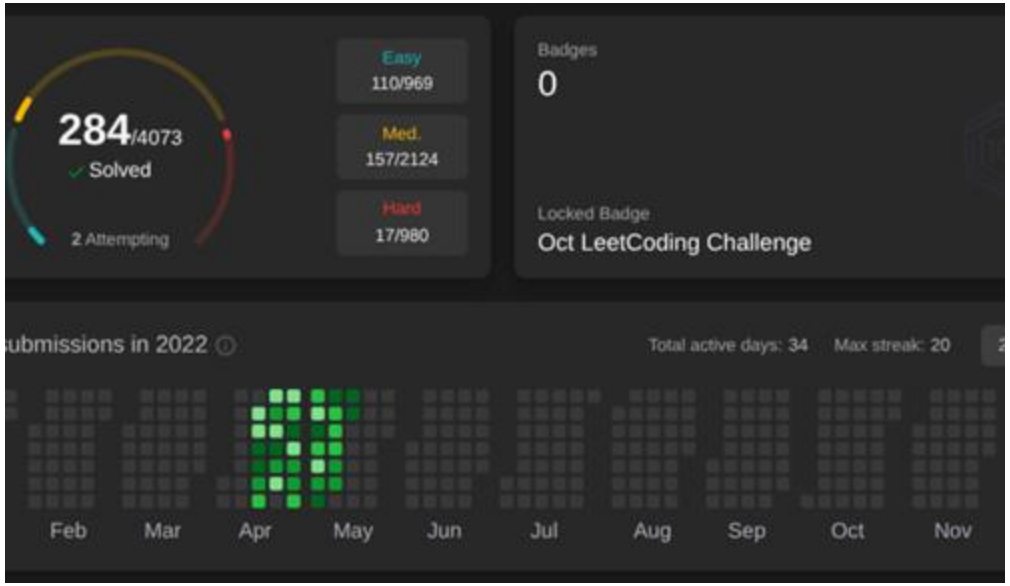
SYSTEMATIC CONSISTENCY

GitHub Contributions & Repos



DELIBERATE PRACTICE

LeetCode Problem Solving



04 | OPEN SOURCE & CURRICULUM

Learn Big Data with AI

A 26-Week Self-Paced Course & Production Data Engineering Guide

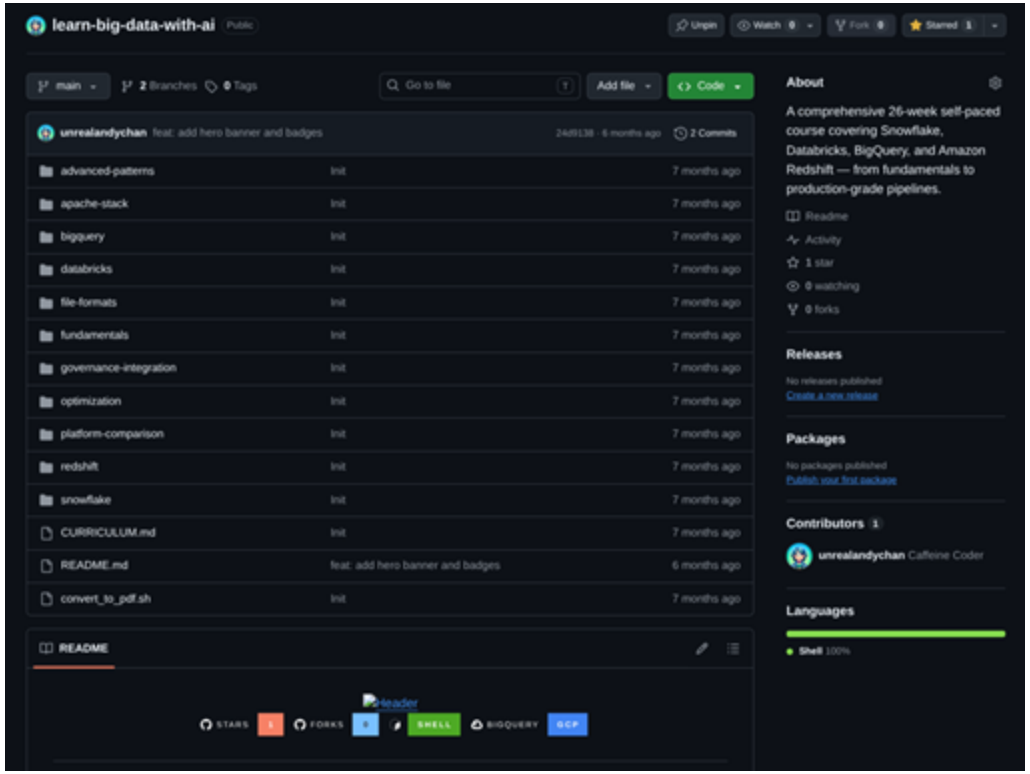
GITHUB REPOSITORY & INTERVIEW PREP

learn-big-data-with-ai

A 26-week self-paced curriculum created for interview preparation, mastering Snowflake, Databricks, BigQuery & Redshift with AI acceleration.

How I Learned & Built with AI:

- **AI-Driven Synthesis:** Used Gemini to structure cross-platform comparative guides
- **Socratic Deep Dives:** Interrogated LLMs on architecture & engine trade-offs
- **Rapid Prototyping:** Scaffolded end-to-end data pipelines for active practice
- **Mock Interrogations:** Drilled complex technical interview questions



04 | STUDY WORKFLOW

My 48-Hour Study Loop with Gemini Notebook

How I Master an Unfamiliar Tech Stack or Client Industry Over a Weekend

STEP 01

Ingest & Map

- Upload docs & repos
- Generate architecture map

STEP 02

Interrogate

- Ask 'Why' & trade-offs
- Quiz yourself with AI

STEP 03

Build Fast

- Scaffold with AI agents
- Code core logic yourself

STEP 04

2-Min Pitch

- Explain business ROI
- Zero jargon allowed

Why I Chose FDE — And Not Management Grade

Why I Refused to Stop Coding in the Middle of the AI Revolution

NOT THIS | MANAGEMENT GRADE

Engineering Manager (EM)

People Politics Over Building

- ✗ **Calendar Tetris** & endless meetings
- ✗ **Org Politics** & review bureaucracy
- ✗ **Technical Edge** quickly goes rusty

MY CHOICE | HIGH-LEVERAGE BUILDER

Forward Deployed Engineer

Hands-On + Executive Impact

- ✓ **100% Hands-On** with frontier AI
- ✓ **1 FDE + AI Agents** = 10-person output
- ✓ **Lead Strategy** without bureaucracy

05 | CAREER CHOICE (2/2)

Why I Chose FDE — And Not Back-Office Core SWE

Owning the Whole Business Puzzle Instead of One Tiny Gear

NOT THIS | BACK-OFFICE SWE

Traditional Core SWE

Deep but Isolated Scope

- ✗ **Narrow Scope:** 1 tiny microservice
- ✗ **Insulated:** 7 layers from customers
- ✗ **Ticket-Driven:** Pre-chewed specs

MY CHOICE | END-TO-END OWNER

Forward Deployed Engineer

Full-Stack + Business Strategy

- ✓ **Own the Whole Puzzle** end-to-end
- ✓ **Direct C-Suite** & user feedback
- ✓ **Uses Both Degrees:** Eng + Business

Should You Try the FDE Path?

An Honest Self-Check for CityU Postgraduate Students

GREAT FIT

Try FDE If You...

High Energy for People + Puzzles

- ✓ **Love Business + Tech** puzzles
- ✓ **Enjoy Talking to People** & coding
- ✓ **Thrive on Fast 0-to-1** building

OTHER PATHS FIT BETTER

Skip FDE If You...

Prefer Predictable Deep Research

- ✗ **Want Fixed Specs** & clean tickets
- ✗ **Dislike Client Meetings** & live demos
- ✗ **Prefer 1 Narrow Topic** for years

06 | ACTION PLAN

How to Start Preparing at CityU Today

3 Practical Steps You Can Take This Semester

ACTION 01

Solve Real Pain

Skip Toy Kaggle Datasets

- Find a real workflow pain
- Deploy a live tool for users

ACTION 02

Master AI Workflows

Gemini Notebook + Agents

- Build your 2nd brain in AI
- Pair-program with agents

ACTION 03

2-Minute Pitch

Speak Business ROI

- Quantify time & cost saved
- Pitch without tech jargon

06 | SUMMARY

3 Key Takeaways from Today

What I Hope You Remember From My Journey

TAKEAWAY 01

Non-CS = Superpower

Curiosity > Credentials

- **Chinese Major → SWE (If I can do it, you also can!)**
- **Starting degree ≠ ceiling**

TAKEAWAY 02

Business × Tech

Syntax Cheap; ROI Scarce

- **Bridge C-Suite & codebase**
- **Keep Learning is the Key to success.**

Stay Hungry, Stay Foolish

TAKEAWAY 03

Stay Hands-On

1 Builder + AI = Squad

- **Learn by doing (Build something fun and verify your idea)**
- **Never stop coding (Even with AI)**

06 | INTERACTIVE Q & A

Q & A — Ask Me Anything!

Thank You! Let's Talk About Your Career & Questions

Career Pivot • Self-Learning SWE • FDE Interviews • Gemini Study Workflow